

Tutorial Worksheet 1 - Extrema of functions and Convexity

Problem 1. (Functions of Two Variables)

- (a) Find the local maximum and minimum of $f(x, y) = 2x^2 - xy + 2y^2 - 20x$.
- (b) Given n points (x_i, y_i) , where the x_i 's are distinct, find a and b such that the function $f(a, b) = \sum_{i=1}^n (ax_i + b - y_i)^2$ is minimum.
- (c) Show that $f(x, y) = y^2 + x^2y + x^4$ has a minimum at $(0, 0)$.
- (d) Show that the function $f(x, y) = 2x^4 - 3x^2y + y^2$ has neither a maximum nor a minimum at $(0, 0)$.
- (e) Find all local maxima and minima of $f(x, y) = xy e^{-x}$.

Problem 2. (Function of Several Variables)

- (a) Find all the local maxima and minima of $f(x, y, z) = x^2 + 2y^2 + z^2 - 2xy + yz$.
- (b) Show that the function $f(x, y, z) = x^4 + y^4 + z^4 - 4xyz$ has a minimum at $(1, 1, 1)$.

Problem 3. (Constrained Optimization with Lagrange Multiplier)

- (a) Find all the local maxima and minima of $f(x, y) = 9 - (x - 1)^2 - y^2$ subject to $x^2 + y^2 = 4$.
- (b) Show that the volume of the greatest rectangular parallelepiped (a three-dimensional geometric shape with six faces, where every face is a parallelogram) that can be inscribed in the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ is $\frac{abc}{3\sqrt{3}}$.
- (c) If $ax^2 + by^2 = ab$, show that the maximum and minimum values of $x^2 + xy + y^2$ are the values of θ given by $4(\theta - a)(\theta - b) = ab$.
- (d) Given m positive numbers $c_1, c_2, \dots, c_m$, find the maximum value of

$$\sum_{i=1}^m c_i x_i,$$

where the variables x_i are restricted so that

$$\sum_{i=1}^m x_i^2 = 1.$$

- (e) Find the minimum value of

$$\sum_{i=1}^n a_i^2$$

subject to

$$\sum_{i=1}^n a_i = 1.$$

- (f) Find the maximum and minimum values of

$$\mathbf{x}^\top A \mathbf{x}$$

subject to

$$\mathbf{x}^\top \mathbf{x} = 1,$$

where A is a symmetric matrix.

(g) If

$$2x + 3y + 4z = a,$$

show that the maximum value of

$$x^2 y^3 z^4$$

is

$$\left(\frac{a}{9}\right)^9.$$

Problem 4. (Geometry of Linear and Quadratic Programming)

(a) Consider the optimization problem

$$\begin{array}{ll} \max_{x,y} & 4x + y \\ \text{s.t.} & 3x + 2y \leq 18, \\ & -x + 2y \geq 2, \\ & x \leq 3, \\ & x, y \geq 0. \end{array}$$

(i) Find the optimal solution when $x, y \in \mathbb{R}$ (Continuous Linear Programming).

(ii) Find the optimal solution when $x, y \in \mathbb{Z}$ (Integer Linear Programming).

(iii) Find the optimal solution when $x \in \mathbb{Z}$, $y \in \mathbb{R}$ (Mixed-Integer Programming).

(iv) Find the optimal solution when $x \in \{0, 1\}$, $y \in \mathbb{R}$ (Binary and Mixed-Integer Decisions).

(b) Consider

$$\begin{array}{ll} \max_{x,y} & 9x^2 + y^2 \\ \text{s.t.} & 3x + 2y \leq 18, \\ & -x + 2y \geq 2, \\ & x \leq 3, \\ & x, y \geq 0. \end{array}$$

Find the optimal value when $x \in \mathbb{R}, y \in \mathbb{R}$ and $x \in \{0, 1\}$, $y \in \mathbb{R}$.

Problem 5. (Convexity)

(a) Let $\mathbf{x} \in \mathbb{R}^n$ and $y \in \mathbb{R}$. Consider the quadratic-over-linear function

$$f(\mathbf{x}, y) = \begin{cases} \frac{\mathbf{x}^\top \mathbf{x}}{y}, & y > 0, \\ +\infty, & y \leq 0. \end{cases}$$

(i) Show that the effective domain of f is convex.

(ii) Compute the gradient $\nabla f(\mathbf{x}, y)$ for $y > 0$.

(iii) Compute the Hessian $\nabla^2 f(\mathbf{x}, y)$.

(iv) Show that the Hessian is positive semidefinite for every $y > 0$.

(v) Hence, conclude that f is a convex function.

(b) Let $\mathbf{x} = (x_1, \dots, x_n)^\top \in \mathbb{R}^n$. Consider the log-sum-exp function

$$f(\mathbf{x}) = \log \left(\sum_{i=1}^n e^{x_i} \right).$$

(i) Determine the domain of f and show that it is convex.

(ii) Compute the gradient $\nabla f(\mathbf{x})$.

(iii) Compute the Hessian $\nabla^2 f(\mathbf{x})$.

(iv) Show that the Hessian is positive semidefinite.

(v) Hence, conclude that f is convex.

(c) Let S_{++}^n denote the set of symmetric positive definite $n \times n$ matrices. Consider

$$f(X) = -\log \det(X), \quad X \in S_{++}^n.$$

(i) Show that S_{++}^n is a convex set.

(ii) Fix $X \in S_{++}^n$ and a symmetric matrix $V \in S^n$. Define the restriction of f to the line $X + tV$ by

$$g(t) = f(X + tV) = -\log \det(X + tV),$$

for all t such that $X + tV \succ 0$.

(iii) Show that

$$X + tV = X^{1/2} \left(I + tX^{-1/2}VX^{-1/2} \right) X^{1/2}.$$

(iv) Let $\lambda_1, \dots, \lambda_n$ be the eigenvalues of

$$M = X^{-1/2}VX^{-1/2}.$$

Show that

$$g(t) = -\log \det X - \sum_{i=1}^n \log(1 + t\lambda_i).$$

(v) Compute $g''(t)$ and show that $g''(t) \geq 0$.

(vi) Hence conclude that

$$\boxed{f(X) = -\log \det X}$$

is convex on S_{++}^n .